scReg-Eval: a fixed-panel audit of regulatory alignment in single-cell RNA foundation-model gene graphs

Zeyu Fu [1]

[1]**State Key Laboratory of Trauma and Chemical Poisoning, Institute of Combined Injury, Chongqing Engineering Research Center for Nanomedicine, College of Preventive Medicine, Army Medical University, Chongqing, China**
[1]**Corresponding author: fuzeyu99@126.com**

ABSTRACT

Background. Claims that single-cell RNA foundation models encode gene regulation are difficult to audit because many reference networks share co-expression structure with the models' training data.
Methods. We introduce **scReg-Eval**, a fixed-panel protocol that compares frozen pairwise foundation-model gene graphs with a sequence- and accessibility-derived regulatory-potential proxy while controlling co-expression, target peak count, gene length, GC content, RNA detection rate, transcription-factor out-degree, and target in-degree. The preregistered panel contains 1,200 genes, including 446 in-range transcription factors (TFs). We evaluate thirteen pooled model/readout combinations across brain and paired peripheral blood mononuclear cell (PBMC) multiome data using two plus-one Monte Carlo randomization tests ($N = 999$ each): a gene-label permutation and a within-TF-row, degree-preserving target shuffle. Benjamini–Hochberg correction is applied within eight prespecified tissue-by-confound-specification-by-null families.
Results. Under the primary full confound specification, seven of thirteen rows are supported by both nulls: six small positive alignments (Geneformer embedding in brain and PBMC, UCE encoder in brain and PBMC, scGPT encoder in brain and PBMC; ρ from 0.00425 to 0.01241) and one negative alignment (Geneformer attention in brain, $\rho = -0.0036$) from the same model whose embedding readout is positive in the same tissue. Two qualifications bound that headline. First, all seven supported rows appear only in the specification that conditions on degree covariates computed from the proxy itself; under the non-degree specification, which carries no such outcome-derived covariates, no row is supported by both nulls, and three of the six positives change sign. Second, the six positive readouts fail the protocol's own preregistered usability check—their FM-versus-co-expression Spearman correlations are negative or near zero (-0.18 to $+0.01$)—so their support is read as readout-specific geometry, not as regulatory recovery. The co-expression baseline, tested on the same edge set, is itself supported in both tissues, and the PBMC Geneformer-embedding alignment ($\rho = 0.00425$) is smaller than its own baseline ($\rho = 0.0070$). A supervised TF-disjoint probe on paired PBMC data gives mixed family-level results: UCE versus co-expression is supported by paired sign-flip testing ($q = 0.0325$), as is the random-init floor ($q = 0.0300$), whereas Geneformer embedding ($q = 0.5125$), Geneformer attention ($q = 0.65$), and scGPT ($q = 0.255$) are not. The probe therefore must not be summarized as showing that all models are indistinguishable from co-expression.
Conclusions. The audit finds weak, readout-specific, specification-dependent alignment concentrated in embedding-geometry readouts, and does not support a global regulatory-capability claim.

Keywords: single-cell RNA-seq, foundation models, gene regulatory networks, co-expression confounding, randomization inference, benchmark audit

1 INTRODUCTION

Single-cell RNA foundation models (scFMs) are increasingly used as frozen representations for network inference and perturbation analysis [24, 7]. A high score against a curated or transcriptome-derived regulatory network, however, need not imply that a model encodes regulation beyond co-expression [20, 16]. Both the reference and the model can inherit gene abundance, detectability, hubness, and co-variation.

We address this problem as an audit-design question. scReg-Eval freezes a transcription-factor (TF)–target panel, constructs an accessibility-and-motif regulatory-potential proxy without expression covariation, applies the same edge mask and confounds to every model, and evaluates the observed alignment against two structurally different randomization nulls. This design does not make the proxy causal ground truth. It asks a narrower, reproducible question: on the fixed panel, does a model gene graph align with the proxy after controlling measured RNA and graph structure, and is that alignment unusual under both label and target-assignment randomizations? Negative benchmarks of foundation-model interpretability under careful controls have precedent [2, 6, 12]; our aim is the audit protocol and its reporting discipline rather than a verdict by itself.

The closest prior result shows that foundation-model attention recovers co-expression rather than unique regulatory signal [13]. Our reference differs from that work's transcriptome-derived networks in construction: the proxy here is built from genomic sequence (motif matches) and chromatin accessibility only, so no RNA covariation enters the reference at all—it cannot be satisfied by the co-expression structure that the model and the reference would otherwise share. Within one model, we additionally find that two readouts of the same frozen network disagree in sign (Geneformer embedding $+0.0044$ versus attention -0.0036 in brain), a per-readout inconsistency that a model-level regulatory account does not predict and that single-readout studies cannot see.

Our contributions are:

1. a hash-pinned 446-TF by 1,200-gene audit panel and a sequence/accessibility regulatory-potential proxy that shows moderate observed agreement across brain, blood, and a mixed fibroblast reprogramming dataset;
2. a full-confound partial-Spearman statistic that controls co-expression and six structural covariates, with a non-degree specification reported as a co-primary sensitivity analysis;
3. two finite-panel Monte Carlo tests with explicit null semantics, deterministic seed streams, plus-one correction, and eight prespecified FDR families;
4. a multi-readout audit of Geneformer, scGPT, scFoundation, and UCE [24, 7, 11, 22, 28] showing small, readout-specific primary alignments—including a sign-inconsistent embedding/attention pair within one model—rather than a uniform positive or negative result; and
5. descriptive per-cell-type, cross-tissue, and axis-aligned injection diagnostics whose scope is kept separate from the pooled inferential families.

scReg-Eval reporting checklist. Freeze and hash the TF–target panel; label the reference as a regulatory-potential proxy rather than causal truth; control co-expression and structural covariates; state exactly what each randomization changes; use plus-one Monte Carlo *p*-values; define FDR families before reading results; and keep exploratory cell-type and sensitivity analyses distinct from the primary test.

2 METHODS

2.1 Regulatory-potential proxy and fixed panel

The panel contains 1,200 genes, including 446 unique in-range transcription factors, frozen by a SHA-256 manifest. Accessible peaks from brain snATAC-seq, paired PBMC multiome, and a chemical reprogramming ATAC dataset (GSE206767, in which day-0 BJ fibroblasts are pooled with day-3 induced neuron-like cells as a single `ALL` accessibility track) are linked to genes by a ± 2 kb promoter-plus-gene-body rule, which retains 6,609 of 111,743 accessible peaks in the PBMC data (5.91%) and systematically excludes distal enhancer elements. JASPAR2024 CORE vertebrate motifs are scanned with MOODS [21, 14]; motif evidence in an accessible peak contributes a directed TF–target regulatory-potential weight, following the motif-in-accessible-region linkage paradigm used by regulon-inference methods [3]. The graph uses no expression covariation, but accessibility, motif occurrence, and proximity do not establish direct regulation or causality. We therefore call it a proxy throughout. The two evaluation tissues differ in pairing: the PBMC data are a paired multiome assay (RNA and ATAC from the same cells), whereas the brain analysis combines snATAC-seq with cross-study RNA, so the co-expression control is intrinsically stronger in PBMC than in brain.

Brain analyses apply a prespecified neural marker mask to every brain readout; in this panel the mask removes 446 of 534,754 brain edges (0.08%), a small fixed-panel exclusion because only one of the 34 prespecified markers (VCAN, which is not a transcription factor) falls in the fixed panel. PBMC edges are unmasked under the prespecified tissue policy. Before analysis, the driver verifies the manifest, TF identity and order across all caches, matrix dimensions and finiteness, and four pinned legacy-file hashes.

2.2 Foundation-model and baseline graphs

Gene graphs are read from cached, model-validated outputs over the same manifest. Available pooled readouts are Geneformer embedding, attention, and in-silico knockout; scFoundation encoder; UCE encoder (4-layer variant, checkpoint SHA-256 pinned by the pipeline); and scGPT encoder; plus a random-init Geneformer floor. Coverage differs by tissue because only precomputed, validated graphs are included: brain has eight FM/readout rows and PBMC has five. Co-expression is reported as a baseline on the same edge set and nulls, but is not included as an FM row or in FM multiplicity families. Co-expression enters its own test only as the signal, never as a covariate: its design matrix is the full specification minus the co-expression column—an intercept plus the six structural covariates (target peak count, gene length, RNA detection rate, GC content, TF out-degree, target in-degree)—because partialling co-expression out of itself is degenerate by construction. CellPLM is outside scope because its architecture does not expose contextual per-gene states [27]. Inference settings differ by model and are reported rather than harmonized: PBMC readouts use a deterministic 4,000-cell pool for every model; brain Geneformer embedding and scGPT use all available cells, brain Geneformer attention is capped at 4,000 cells, brain UCE uses 2,000 cells, and brain scFoundation uses 1,500 cells with a 512-token context cap. Manifest gene coverage is 1,159/1,200 for PBMC UCE, 1,195/1,200 for brain UCE, 1,199/1,200 for brain scFoundation, 1,168/1,200 for PBMC scFoundation, and 1,200/1,200 otherwise.

2.3 Observed statistic and confound specifications

For each retained TF–target pair, all graph weights are rank transformed. The primary statistic is the Pearson correlation between residualized ranks, equivalent to partial Spearman correlation. The *full* design contains an intercept, ranked co-expression, and standardized target peak count, gene length, RNA detection rate, GC content, TF out-degree, and target in-degree. Peak-count and GC covariates are computed from at most 40 peaks per gene, so they understate the extremes of very peak-dense genes. The two degree covariates are computed from the proxy graph itself, so the full specification conditions on functions of the outcome; this is a validity hazard, not a bookkeeping detail, and it is why the *non-degree* design, which omits both degree columns, is reported co-primary rather than as a footnote. Observed and randomized statistics always use matching column sets.

2.4 Finite-panel randomization inference

The TF panel is treated as fixed; no TF-superpopulation bootstrap [8] or confidence interval is used. For the gene-label null, one permutation relabels both endpoints of the proxy graph, in the spirit of the Mantel matrix correspondence test [15]. The FM graph, co-expression, and non-degree gene covariates remain fixed. In the full specification, proxy-derived TF out-degree and target in-degree are recomputed after relabeling.

The second null independently shuffles edge weights within each TF row across non-self targets. This preserves TF out-degree and the row's weight multiset while changing target assignment; target in-degree is recomputed only for the full specification. One proxy randomization per replicate is shared across model rows, with each model scored against the same replicate. Explicit-index tests verify numerical equivalence to separate least-squares calculations, including rank-deficient designs.

Both tests use two-sided plus-one correction [18],

$$p_{\mathrm{MC}} = \frac{1+\sum_{b=1}^{999} \mathbf{1}\{|T_b| \geq |T_{\mathrm{obs}}|\}}{1000},$$

with resolution 0.001. Benjamini–Hochberg correction [4] is performed within each of eight families: tissue (brain/PBMC) $\times$ confound specification (full/non-degree) $\times$ null (gene-label/row-shuffle). The full specification is the preregistered primary; the non-degree specification is reported alongside it as co-primary because the full specification conditions on proxy-derived degree covariates.

2.5 TF-disjoint supervised probe

The pooled audit is an unsupervised raw-readout test. As a supervised complement, run on paired PBMC multiome data only, we train a ridge probe to predict each TF's proxy-target profile from the FM graph itself. The probe was originally specified over per-gene embeddings; because the validated caches store gene-by-gene graphs rather than embeddings, we reformulated it at the pair level, which answers the same question without new model inference. Samples are ordered (TF, gene) pairs; features for each family are the edge weight $S_{tf,g}$, its reverse $S_{g,tf}$, and the within-TF rank of the edge. These are scalar per-edge summaries, so the probe tests edge-level linear recoverability, not row-level regulatory programs. Nothing in the design is indexed by TF identity, so a probe trained on 311 held-in TFs transfers unchanged to 135 disjoint test TFs. Regularization is chosen by leave-one-out cross-validation over training pairs. Performance is the per-test-TF Spearman correlation between predicted and observed proxy profiles, averaged over test TFs, after residualizing both sides against target peak count, gene length, GC content, and detection rate. The null permutes gene labels within each test TF; paired FM-versus-co-expression contrasts use sign-flipped per-TF differences. Gene-level degree features are withheld from the primary probe arm because they let the probe reach the proxy through gene-level confounding rather than through the graph's edges.

2.6 Descriptive and pipeline-sensitivity analyses

Per-cell-type rows report observed partial correlations, cell counts, ranges, medians, and sign counts. They have no randomization p-values or FDR adjustment. Cross-tissue proxy reproducibility is also observed-only. Finally, for each tissue we evaluate the following injected fractions:

$$\alpha \in \{0, .02, .05, .08, .12, .18, .25, .35, .5, .75, 1\}.$$

For each value, we combine an independently generated noise vector with an α fraction of the proxy residual and repeat the observed statistic 30 times. This axis-aligned diagnostic checks implementation response; it is not a power analysis and does not define an MDE, confidence interval, or exclusion bound.

The fixed panel, model/readout availability, inference cell counts, and edge-accounting steps are audited explicitly in Fig. 1; blank coverage cells denote missing validated graphs, not negative results. The full audit therefore comprises 13 pooled FM/readout rows rather than a balanced model-by-tissue factorial design.

3 RESULTS

3.1 The proxy is reproducible, but remains a proxy

The accessibility/motif construct shows moderate observed agreement across tissue consensus graphs: brain–PBMC $\rho = 0.5249$, brain–fibroblast-mix $\rho = 0.5245$, and PBMC–fibroblast-mix $\rho = 0.4623$ (Fig. 2; Table 1). Table 1 also reports edge Jaccard, mean per-TF-row Spearman, residual agreement after additive marginal fits, and binary support φ for the same pairs. The third dataset (GSE206767) is a chemical-reprogramming experiment whose accessibility track pools day-0 fibroblasts and day-3 induced neuron-like cells, so it does not represent any resting tissue. These are descriptive fixed-panel correlations without new randomization inference. They show the construct is stable across ATAC inputs of very different provenance, not a causal ceiling or proof of TF–target truth. Figure 2 also makes the proxy's motif-noise and panel- composition limits explicit. Most of that stability is marginal structure: an additive TF-row-plus-gene-column model fit on one tissue's proxy predicts the other tissue's edges at $\rho = 0.32$–0.41 (69–78% of the observed agreement; Fig. 3), and after removing each tissue's own additive structure the residual agreement is $\rho = 0.41$–0.55. One pair deviates from that monotone reading: for brain–fibroblast-mix the additive fit is net-subtractive (residual $\rho = 0.55$ above the observed 0.52), so its additive-prediction fraction quantifies similarity in the marginals rather than decomposing the edge-level agreement. The dominant share of the apparent reproducibility is therefore fixed by the motif library and the proximity rule, both of which are tissue-independent by design, rather than by tissue-specific regulatory biology.

3.2 Primary full-confound audit: six weak positives, one negative, and a baseline that matches them

Table 2 and Fig. 4 report all thirteen pooled FM/readout rows plus the co-expression baseline. Six small positive effects are supported by both randomization tests after within-family BH correction. In brain,

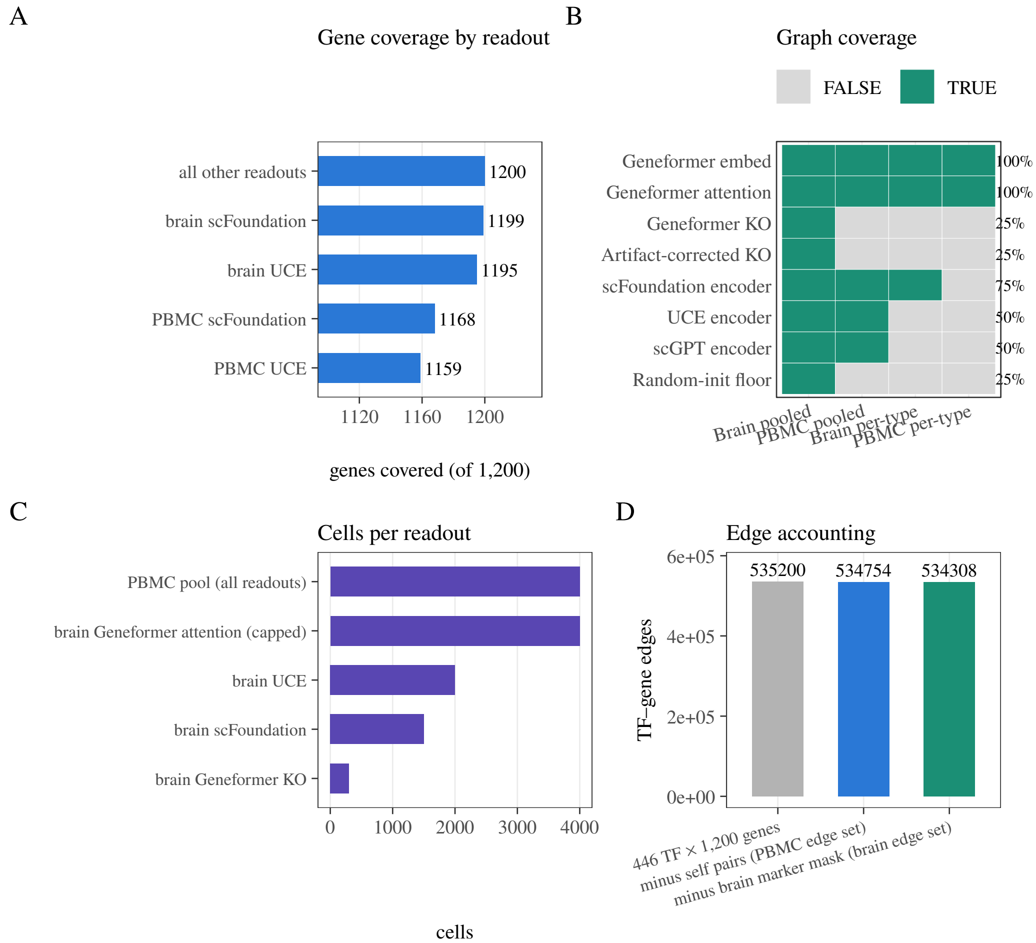

Figure 1. Coverage and fixed-panel quality control. (A) Manifest-gene coverage for encoder readouts. (B) Validated graph availability by tissue and analysis level; coverage is intentionally reported rather than assumed balanced. (C) Cell counts used by representative pooled inference runs. (D) Edge accounting from the $446 \times 1{,}200$ Cartesian panel through self-pair and brain marker-mask exclusions.

Tissue pair	Obs. ρ	Add. frac.	Resid. ρ	φ	Edge Jac.	Mean row ρ	Shared TFs
Brain–PBMC	0.5249	0.7543	0.5073	0.5070	0.416	0.4001	446
Brain–Fibroblast mix	0.5245	0.7771	0.5502	0.5004	0.414	0.4151	446
PBMC–Fibroblast mix	0.4623	0.6895	0.4117	0.4461	0.374	0.3364	446

Table 1. Descriptive cross-tissue reproducibility of the regulatory-potential proxy on the fixed panel. Columns report observed Spearman ρ, the fraction of that agreement explained by TF-row and gene-column additive marginals, residual Spearman after each tissue's own additive fit, binary support φ, edge Jaccard, mean per-TF-row Spearman, and the number of shared TFs. These are fixed-panel descriptive quantities with no confidence interval and no new randomization inference.

Geneformer embedding has $\rho = 0.00442$ ($q_M = 0.006$, $q_D = 0.003$), UCE encoder has $\rho = 0.00461$ ($q_M = 0.004$, $q_D = 0.003$), and scGPT encoder has $\rho = 0.01241$ ($q_M = 0.004$, $q_D = 0.003$). In paired PBMC, Geneformer embedding has $\rho = 0.00425$ ($q_M = 0.002$, $q_D = 0.003$), scGPT encoder has $\rho = 0.01098$ ($q_M = 0.002$, $q_D = 0.003$), and UCE encoder has $\rho = 0.00593$ ($q_M = 0.002$, $q_D = 0.003$).

One further row is supported by both nulls with the opposite sign: brain Geneformer attention, $\rho = -0.00362$ ($q_M = 0.005$, $q_D = 0.016$), from the same model whose embedding readout is positive in the same tissue. A sign-inconsistent pair of readouts within one model is not what a regulatory-capability account predicts.

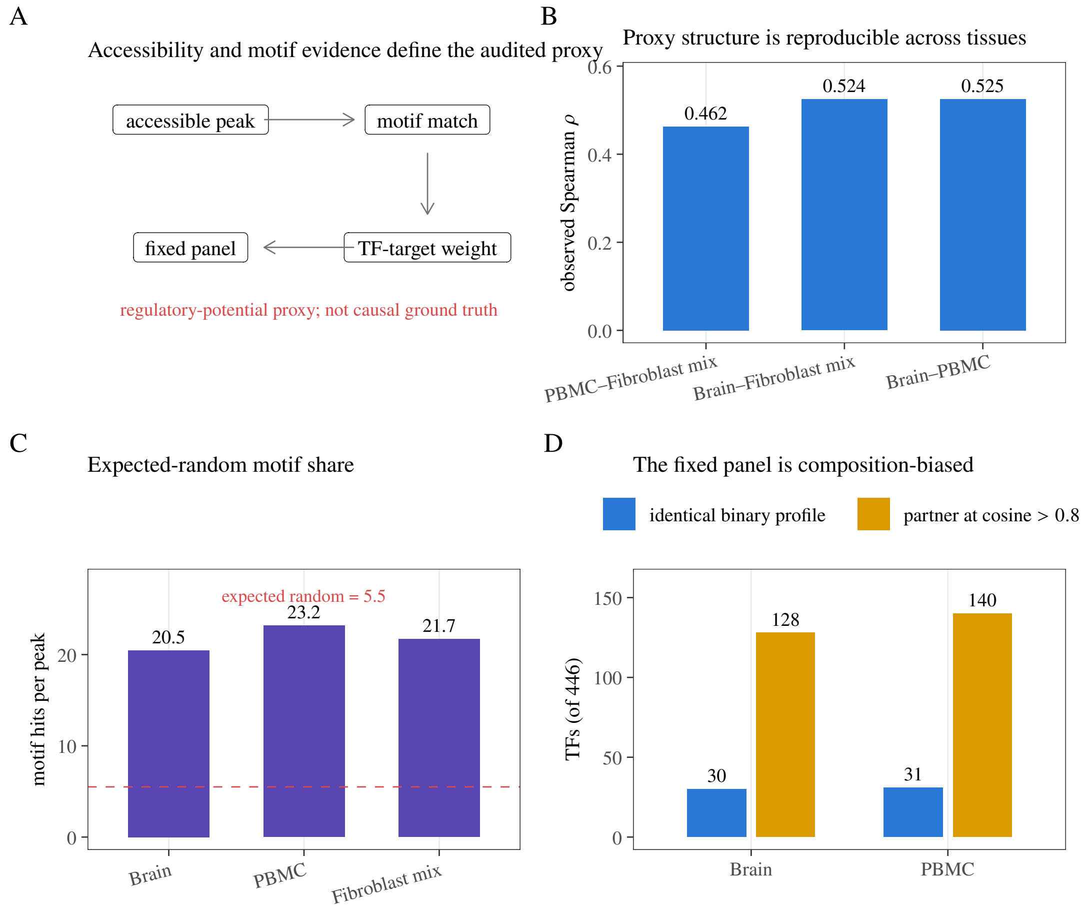

Figure 2. Regulatory-potential proxy, evidence, and panel composition. (A) Construction from accessible peaks, motif matches, and TF–target linkage to the fixed panel; this remains a proxy, not causal ground truth. (B) Observed consensus-graph correlations across three tissue datasets. (C) Motif hits per accessible peak relative to the expected 5.5 random hits at the 10^{-5} motif threshold. (D) TF target-profile redundancy: counts with an identical binary profile or a partner at cosine similarity > 0.8, out of 446 TFs.

The remaining rows are not supported: brain scFoundation ($\rho = 0.00062$), Geneformer knockout (-0.00093) and its artifact-corrected knockout readout (-0.00189; the artifact-corrected row subtracts a random-token position-shift baseline and is not a positive control—the genuine positive control in this audit is the signal-injection diagnostic below), the random-init Geneformer floor (0.00187), and PBMC Geneformer attention (-0.00018) and scFoundation (0.00112).

The co-expression baseline, run through the same nulls on the same edge set (it cannot enter the FM families because partialling co-expression out of itself is degenerate), is itself supported in both tissues: brain $\rho = 0.00276$ ($p_M = 0.019$, $p_D = 0.014$) and PBMC $\rho = 0.00697$ ($p_M = 0.001$, $p_D = 0.001$). Two observations follow. First, the baseline reaches the same order of alignment as the supported FM rows. Second, the PBMC Geneformer-embedding positive ($\rho = 0.00425$) is smaller than its own baseline ($\rho = 0.00697$): on that row the model aligns with the proxy less than the confound it is adjusted against.

Four completed enhancement-v1 checks qualify the fixed-panel interpretation without changing the frozen primary families. Five-fold cross-fitting of the proxy-derived degree covariates preserved the qualitative direction in all 13 pooled rows, with absolute changes from the frozen estimates no larger than 1.1×10^{-5} (E1). Linkage sensitivity preserved the edge-weight structure for TSS–1 kb and TSS–5 kb policies ($\rho = 0.988$ and 0.971 versus the baseline), whereas a promoter-only policy changed the construct substantially ($\rho = 0.119$; E2). In 20 fixed 80% TF subsamples, signs were concordant in all brain and PBMC draws; the descriptive ρ ranges were 0.00271–0.00610 and 0.00115–0.00716, respectively (E3). Detection-based stratification also showed larger alignment for the operational housekeeping proxy than

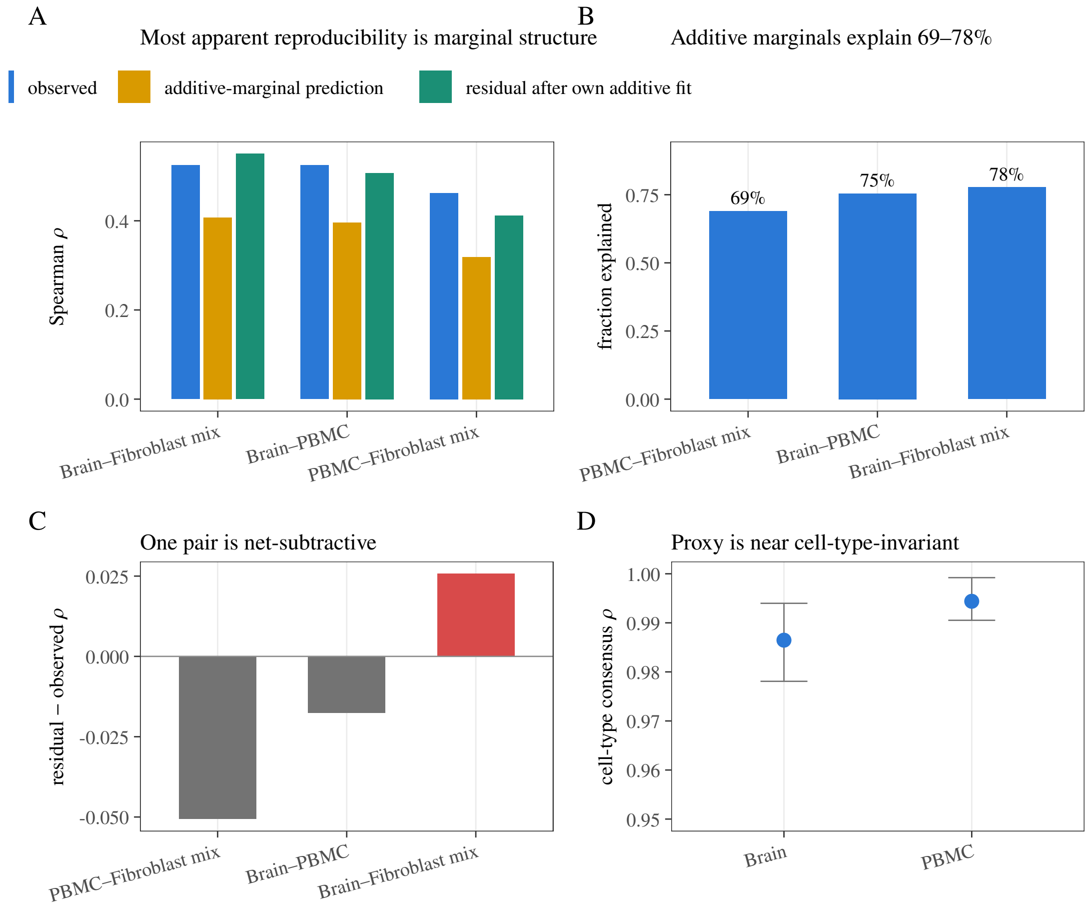

Figure 3. Cross-tissue structure of the regulatory-potential proxy. (A) Observed agreement, TF-row-plus-gene-column additive prediction, and agreement after removing each tissue's own additive fit. (B) Fraction of observed agreement predicted by additive marginals. (C) Difference between residual and observed correlation; brain–fibroblast mix is net-subtractive. (D) Range and mean of pairwise cell-type proxy correlations within brain and PBMC. These are descriptive fixed-panel quantities without new randomization inference.

for low-detection TF strata (brain $\rho = 0.00891$ versus 0.00398; PBMC 0.02111 versus 0.00141; E5). These checks report observed stability and construct sensitivity; E3 did not recompute valid randomization nulls after TF subsampling, and E5 uses a detection-based operational label rather than a biological housekeeping annotation.

A separate sequence-opportunity audit (S1) applied a five-bin target-opportunity permutation and passed its null-calibration checks, but it is not the frozen v2 primary result: several row-level statistics and q-values differ under that stricter sensitivity design. We therefore report S1 as a sensitivity audit rather than claiming that the primary headline has been re-tested under S1. External binding calibration against ENCODE-compatible edges remains approval-blocked and is not treated as completed evidence. Our protocol pre-registers a readout usability check in which a usable readout should correlate with co-expression (attention $\approx$ co-expression is well established). The six supported positive rows come from readouts that fail that check: their FM–co-expression Spearman correlations are -0.18 to $+0.01$ (all but one negative; PBMC scGPT is $+0.01$). The readouts that pass it (scFoundation $+0.08$, PBMC Geneformer attention $+0.40$) are null. Embedding-cosine anisotropy interacting with degree conditioning is a more parsimonious account of that inversion than regulatory encoding, so we report the six positives as readout-specific geometry rather than as regulatory signal (Fig. 5).

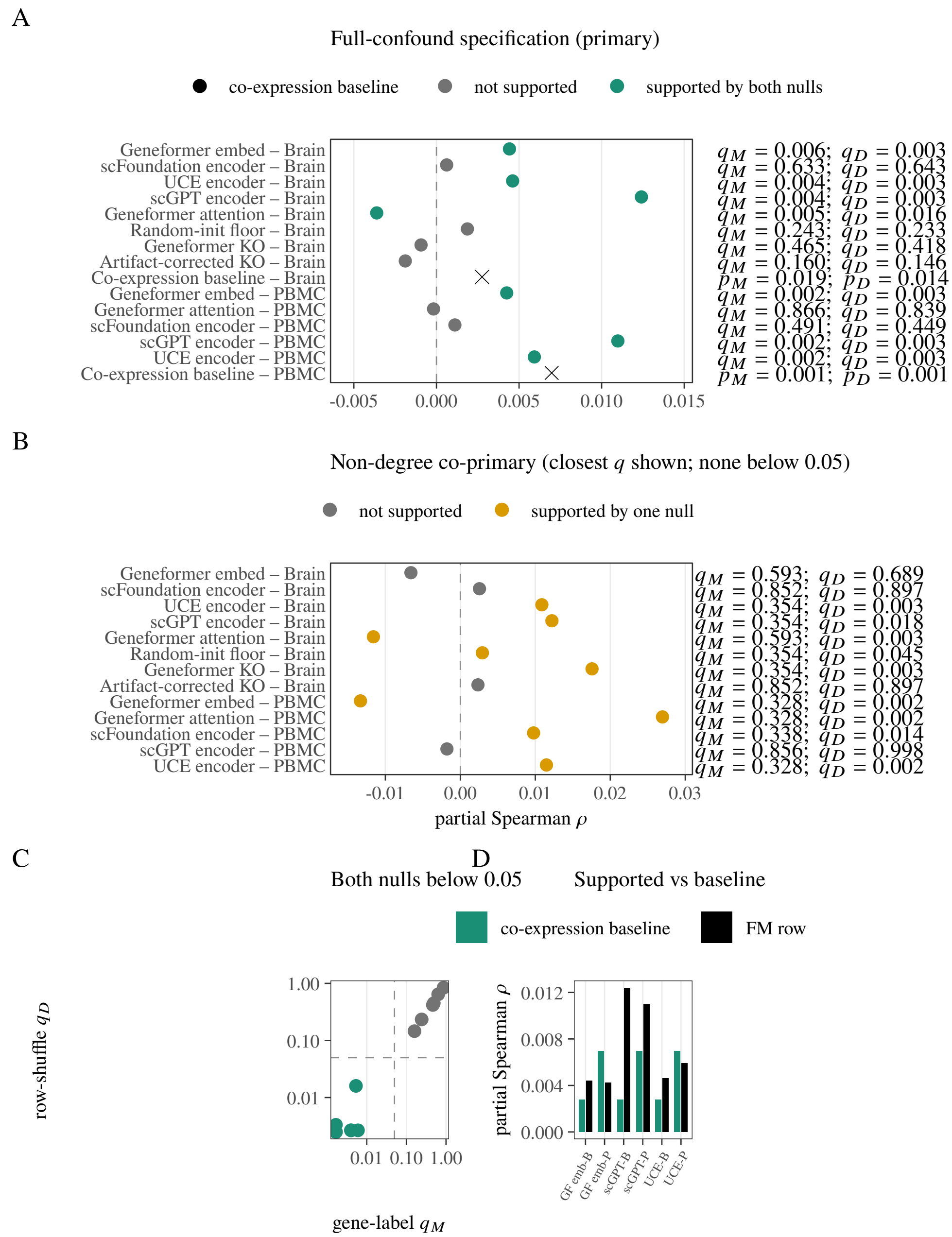

Figure 4. Primary fixed-panel audit under both prespecified confound specifications. (A) Full- confound pooled rows and the same-edge co-expression baseline; labels give BH-adjusted q_M and q_D for FM rows and uncorrected p_M, p_D for the baseline. (B) Non-degree co-primary rows; no row is supported by both nulls. (C) Full-confound q_M versus q_D; support requires both coordinates below 0.05. (D) Each supported positive row compared with its own-tissue co-expression baseline.

Tissue	Readout	Partial ρ	p_M	q_M	p_D	q_D	Support
Brain	Geneformer embed	0.00442	0.003	0.006	< 0.001	0.003	both
Brain	scFoundation encoder	0.00062	0.633	0.633	0.643	0.643	neither
Brain	UCE encoder	0.00461	< 0.001	0.004	< 0.001	0.003	both
Brain	scGPT encoder	0.01241	< 0.001	0.004	< 0.001	0.003	both
Brain	Geneformer attention	-0.00362	0.002	0.005	0.008	0.016	both (neg)
Brain	Random-init floor	0.00187	0.182	0.243	0.175	0.233	neither
Brain	Geneformer KO	-0.00093	0.407	0.465	0.366	0.418	neither
Brain	Artifact-corrected KO	-0.00189	0.100	0.160	0.091	0.146	neither
Brain	Co-expression baseline	0.0028	0.019	–	0.014	–	–
PBMC	Geneformer embed	0.00425	< 0.001	0.002	0.002	0.003	both
PBMC	Geneformer attention	-0.00018	0.866	0.866	0.839	0.839	neither
PBMC	scFoundation encoder	0.00112	0.393	0.491	0.359	0.449	neither
PBMC	scGPT encoder	0.01098	< 0.001	0.002	< 0.001	0.003	both
PBMC	UCE encoder	0.00593	< 0.001	0.002	< 0.001	0.003	both
PBMC	Co-expression baseline	0.0070	< 0.001	–	< 0.001	–	–

Table 2. Primary fixed-panel results (full confound specification). M denotes the gene-label null and D the within-TF-row degree-preserving target shuffle; each q is BH-adjusted within its tissue-by-specification-by-null family. Support is *both* when $q_M < 0.05$ and $q_D < 0.05$, *both (neg)* when that dual support occurs with negative partial ρ, and *neither* otherwise. The co-expression baseline is tested on the same edge set outside the FM families; its entries are uncorrected p-values (dashes mark where no family correction or Support label applies).

Tissue	Confound spec	n rows	$\rho_{\min}$	$\rho_{\max}$	ρ_{median}
Brain	full	15	-0.00396	0.00445	0.00258
Brain	non-degree	15	-0.00986	0.01625	-0.00053
PBMC	full	14	0.00222	0.00682	0.00413
PBMC	non-degree	14	-0.01303	0.05371	0.01502

Table 3. Per-cell-type descriptive ranges of observed partial ρ under full and non-degree confound specifications (brain: 15 exploratory FM$\times$type rows; PBMC: 14). Min, max, and median are taken across those rows only. No randomization p-values, BH correction, or population-uncertainty interpretation.

3.3 The alignments depend on conditioning on proxy-derived degree covariates

Removing the two degree covariates changes both effect direction and randomization behavior (Fig. 7). No non-degree row is supported by both nulls, and three of the six full-spec positives (brain Geneformer embedding, PBMC Geneformer embedding, PBMC scGPT) are negative before degree conditioning. Because the degree covariates are computed from the proxy graph itself, the full specification conditions on functions of the outcome; the positives therefore appear in exactly the specification that carries this hazard. We report the non-degree specification as co-primary and treat the divergence as a substantive finding, not a footnote: the row-shuffle null alone supports several non-degree rows while the gene-label null supports none, so degree adjustment and null choice define different finite-panel questions.

3.4 Per-cell-type patterns are descriptive

The proxy itself is near cell-type-invariant on this panel (pairwise consensus $\rho \approx 0.99$), so per-cell-type rows score readouts against a reference with essentially no cell-type contrast; we report them only as descriptive robustness summaries. The full-confound cell-type analysis contains 15 brain (3 readouts $\times$ 5 cell types) and 14 PBMC (2 readouts $\times$ 7 cell types) FM rows per specification (Fig. 8; Table 3). Brain full-confound estimates range from -0.00396 to 0.00445 with median 0.00258; PBMC estimates range from 0.00222 to 0.00682 with median 0.00413. Under the non-degree specification the same exploratory rows spread further (brain median -0.00053, range $[-0.00986, 0.01625]$; PBMC median 0.01502, range $[-0.01303, 0.05371]$; Table 3). These rows have no randomization p-values or BH correction. They neither support cell-type discoveries nor provide population uncertainty.

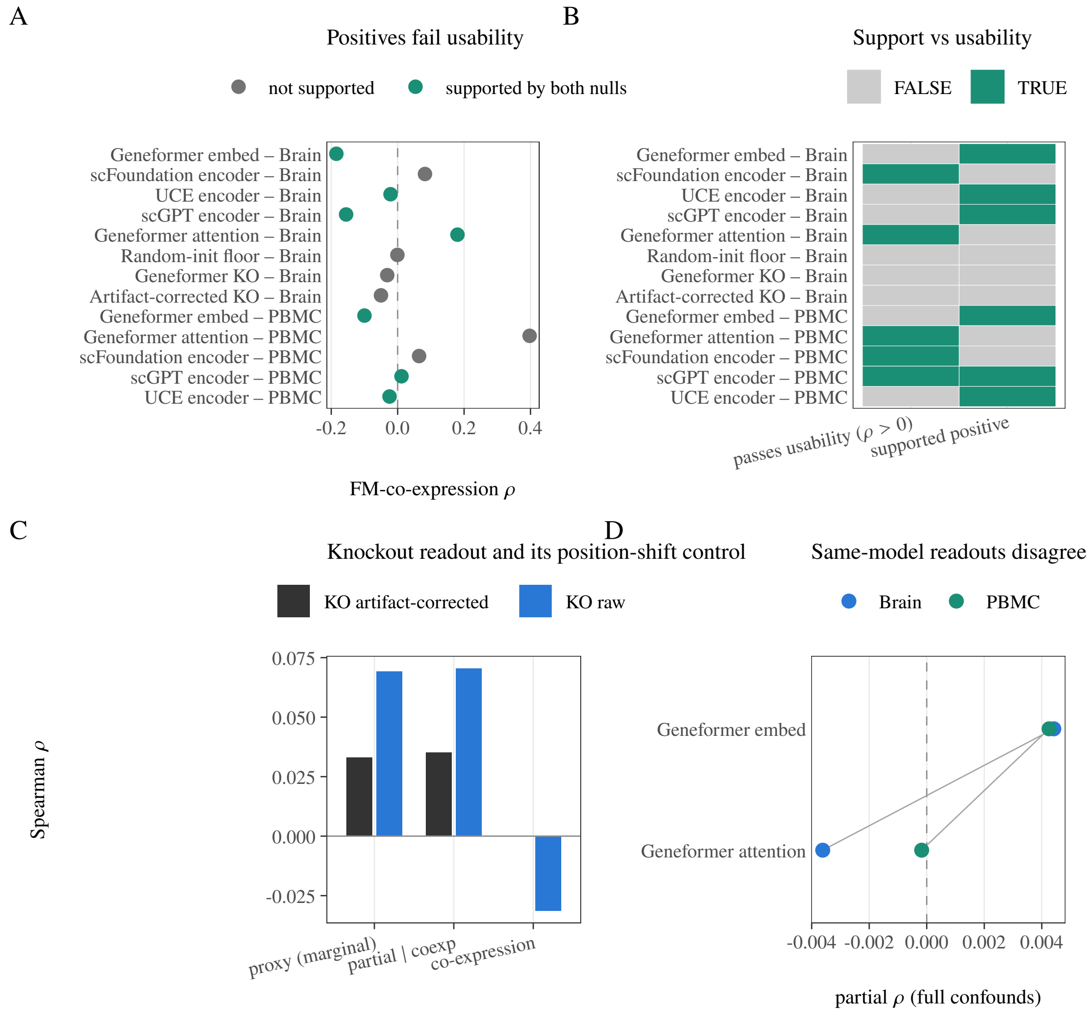

Figure 5. Readout usability and internal consistency. (A) FM–co-expression correlation for all 13 pooled readouts, colored by full-confound support. (B) Positive joint-null support and positive usability shown as separate binary checks. (C) Marginal and partial behavior of the Geneformer knockout readout and its artifact-corrected knockout readout. (D) Geneformer embedding and attention effects within each tissue; the brain effects have opposite signs.

3.5 The implementation responds monotonically to aligned injection, and the observed effects sit at the bottom of that ladder

The axis-aligned diagnostic rises monotonically from approximately zero at $\alpha = 0$ to mean partial correlations of 0.905 in brain and 0.913 in PBMC at $\alpha = 1$ (Fig. 9). The original ladder probes $\alpha \in \{0, .02, .05, .08, .12, .18, .25, .35, .5, .75, 1\}$; the later $\alpha \in \{0.002, 0.005, 0.01\}$ points are subdivided diagnostics used only to locate the observed magnitudes on that curve. They are not direct tests at each row's inferred equivalent and do not constitute a power analysis, MDE, confidence interval, or exclusion bound.

3.6 A supervised TF-disjoint probe gives mixed family-level results

A linear probe trained to predict each TF's proxy targets from graph features on 311 held-in TFs and tested on 135 disjoint TFs (Fig. 10) gives mixed family-level results after confound adjustment. Paired sign-flip testing supports the UCE-versus-co-expression contrast ($q = 0.0325$) and the random-init floor contrast ($q = 0.0300$), while the Geneformer embedding ($q = 0.5125$), Geneformer attention ($q = 0.65$), and scGPT ($q = 0.255$) contrasts are not supported. Two properties of that test qualify the outcome. First, the Benjamini–Hochberg family contains the random-init floor—a methodological control rather than an FM readout—so that families are compared against the baseline on equal footing; excluding it

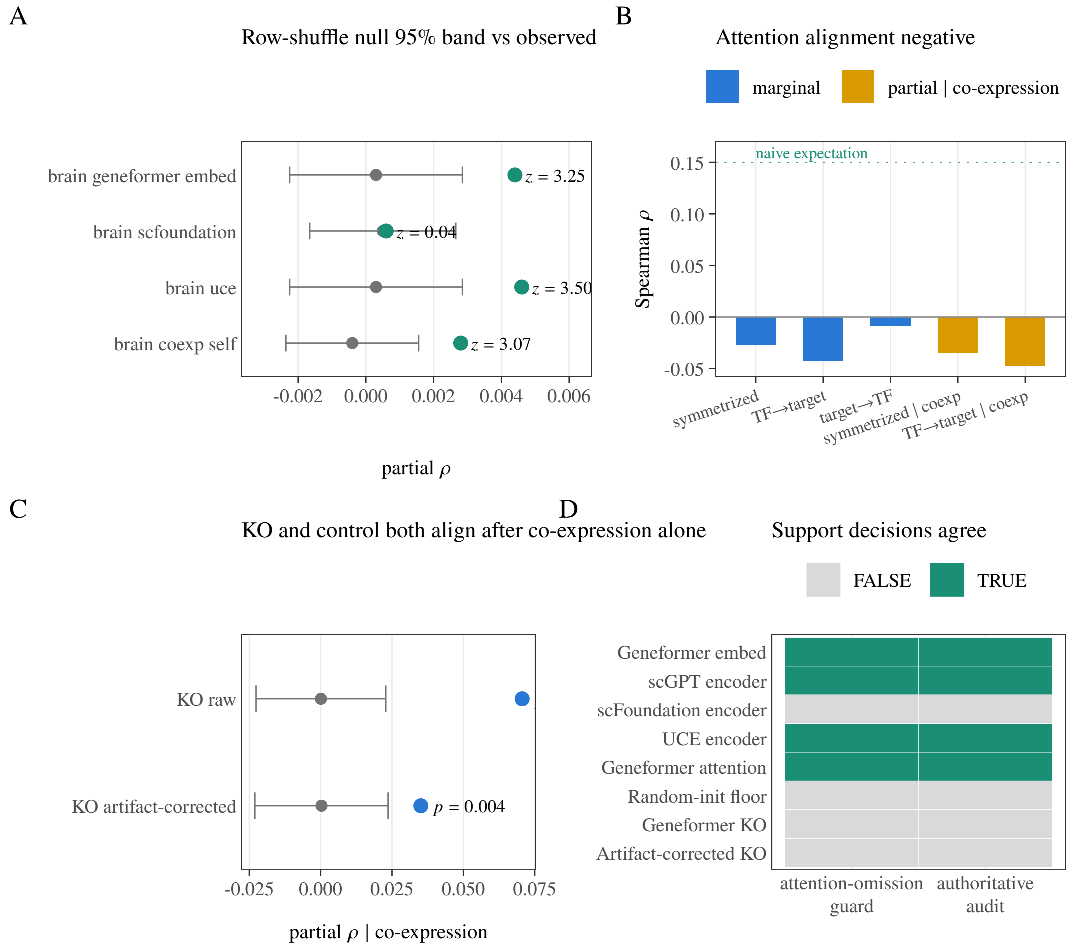

Figure 6. Randomization and readout diagnostics. (A) Observed statistics against the row-shuffle null mean and 95% null band for representative rows. (B) Brain Geneformer-attention marginal and co-expression-adjusted directions. (C) Geneformer knockout and artifact-corrected knockout both align with the proxy after partialling co-expression alone ($p = 0.001$ and $p = 0.004$); the audit's full confound specification removes that alignment (Table 2). (D) Joint-null support decisions are identical between the authoritative audit and the attention-omission guard.

($n = 4$) raises the UCE contrast to $q = 0.052$, just above the 0.05 line. Second, the sign-flip statistic here compares how much each family is over-corrected, not how much it recovers: adjusted test correlations are essentially zero for UCE ($+0.0003$) and the floor ($+0.0023$) but negative for co-expression (-0.0120), so the supported contrasts mean the baseline is over-corrected further below zero, not that UCE carries more signal. The adjusted test correlations span -0.019 to $+0.002$ overall. The probe's confound set differs from the audit's—it contains neither co-expression nor the proxy-derived degree covariates—so it answers a different question (supervised recoverability, not raw-readout alignment) and neither confirms nor refutes the pooled rows. In particular, the random-init result prevents interpreting the UCE contrast as model-specific regulatory recovery, while the UCE result prevents a blanket claim that every FM is indistinguishable from co-expression.

3.7 The construct transfers across three independent ATAC datasets

The proxy was built independently in three datasets (brain GSE174367, PBMC 10x multiome, and the GSE206767 fibroblast reprogramming pool), and Fig. 11 quantifies how much of each dataset the construct actually uses and shares. The ± 2 kb linkage admits only a small fraction of each dataset's peaks: 8,823 of 219,070 in brain (4.03%), 6,609 of 111,743 in PBMC (5.91%), and 11,614 of 275,448 in the fibroblast pool (4.22%)—the proximity-linkage limitation is therefore a three-tissue property, not a PBMC-specific one. On the shared non-self TF–gene edge set (534,754 edges), 38,082 edges carry

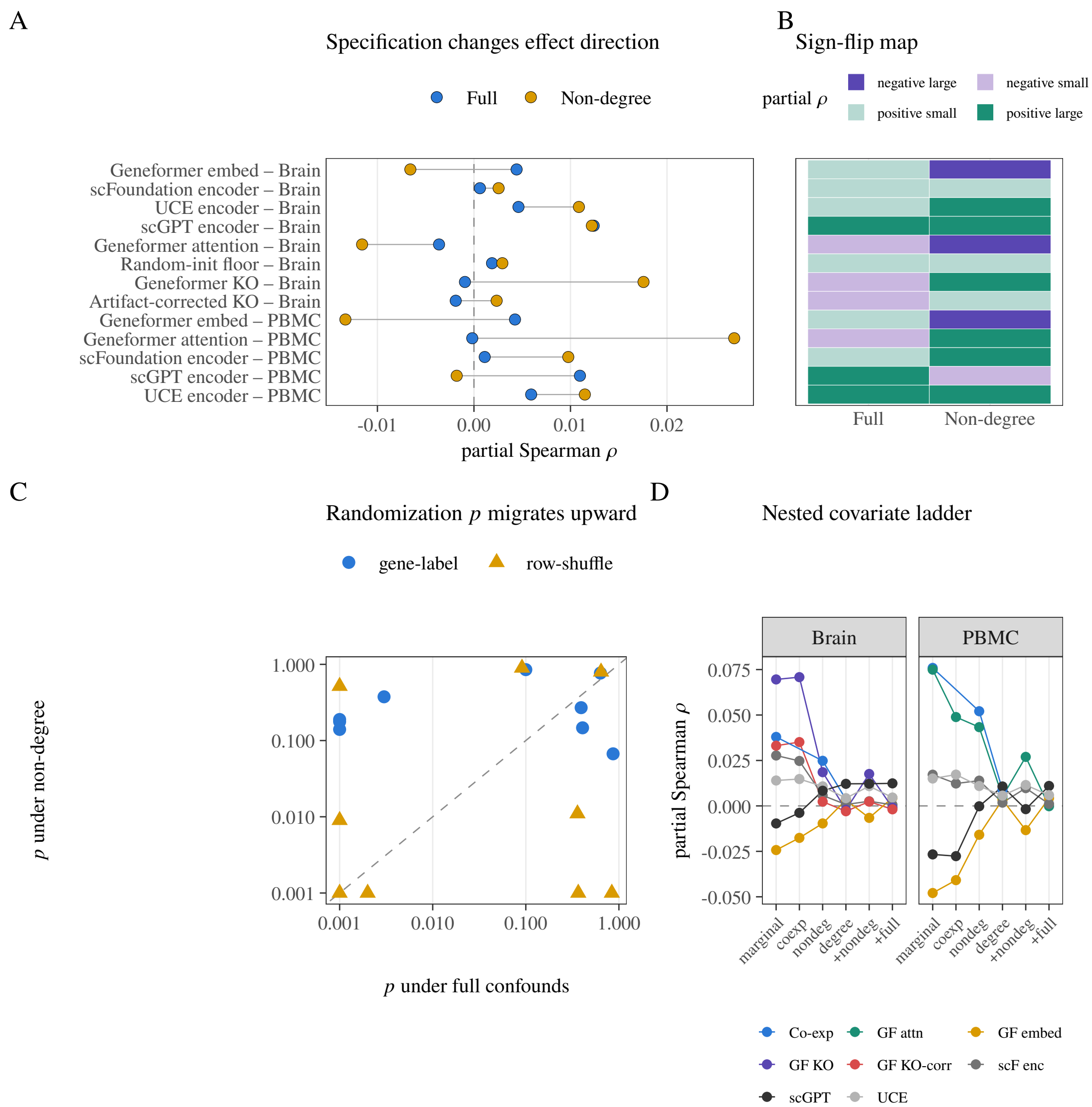

Figure 7. Dependence on the confound specification. (A) Full and non-degree estimates joined within each row. (B) Effect-sign map across specifications. (C) Full-confound versus non-degree randomization p-values for both nulls. (D) Nested covariate ladder from marginal correlation to the primary rung; lines are descriptive and are not a model-selection exercise.

nonzero weight in all three consensus proxies (24% of the 158,838-edge union), and the TF out-degree marginals are similar across tissues (median 129–171 supported targets per TF)—which is exactly the structure the additive decomposition of Fig. 3 attributes most cross-tissue agreement to. Cross-tissue rank agreement also tracks binary-support agreement (observed ρ 0.462–0.525 versus support φ 0.446–0.507), so the construct transfers chiefly at the level of which edges exist, not of their continuous weights. The fibroblast arm carries no foundation-model readouts and enters none of the audit's supported rows; it is a construct-level check, not evidence about model capability.

4 DISCUSSION

The audit finds six small positive fixed-panel alignments and one negative one, all supported by both randomization nulls under the full confound specification. Three features of that result restrain its interpretation. First, the co-expression baseline, tested on the same edge set, is itself supported in both tissues, and one of the six positives (PBMC Geneformer embedding) is smaller than its own baseline; the

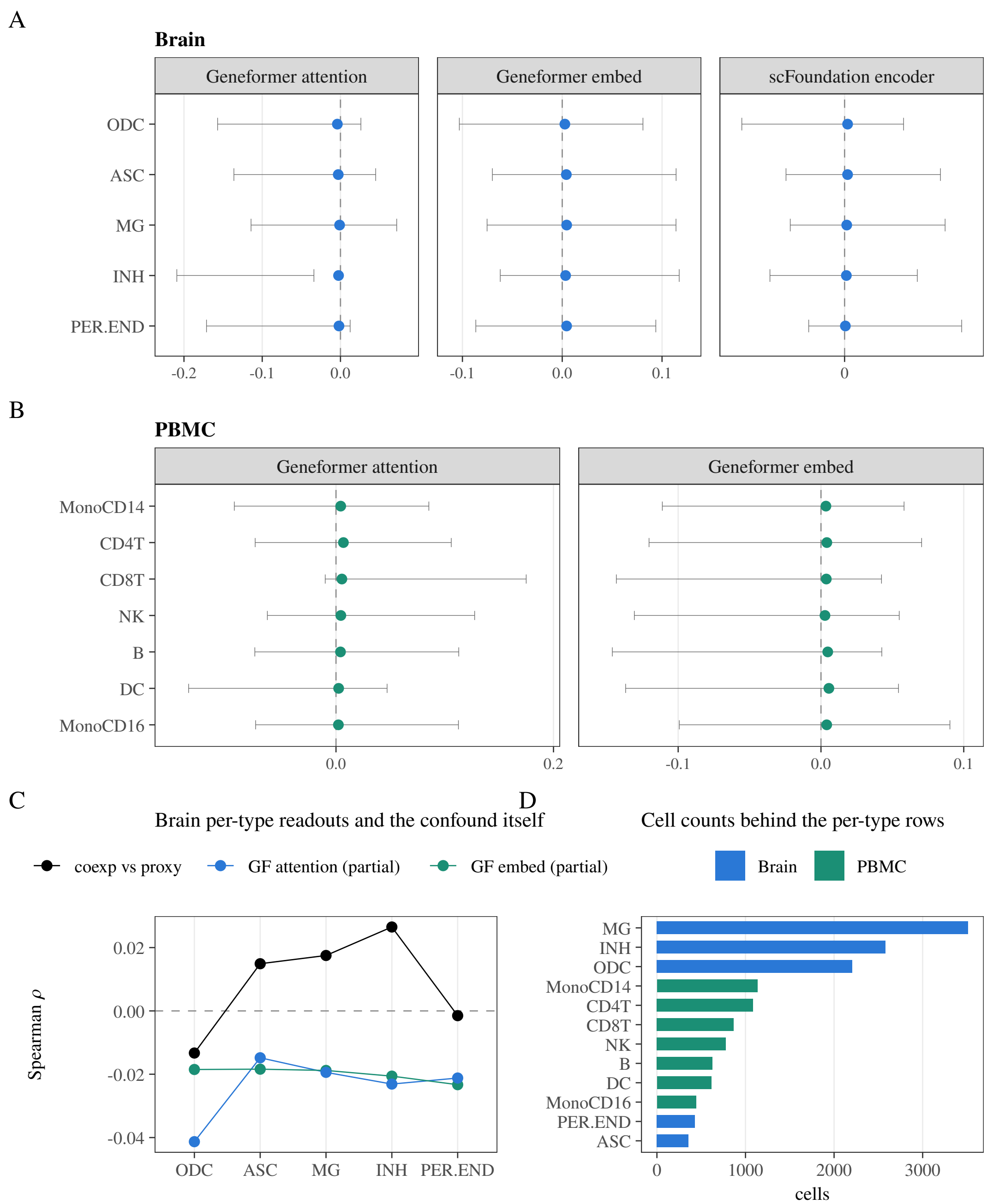

Figure 8. Per-cell-type descriptive robustness summaries. (A) Brain full-confound rows by readout. (B) PBMC full-confound rows by readout. (C) Earlier brain per-type co-expression, Geneformer- embedding, and Geneformer-attention summaries on the same fixed panel. (D) Cell counts underlying the per-type rows. These observed effects have no randomization p-values, multiplicity claims, or population-uncertainty interpretation.

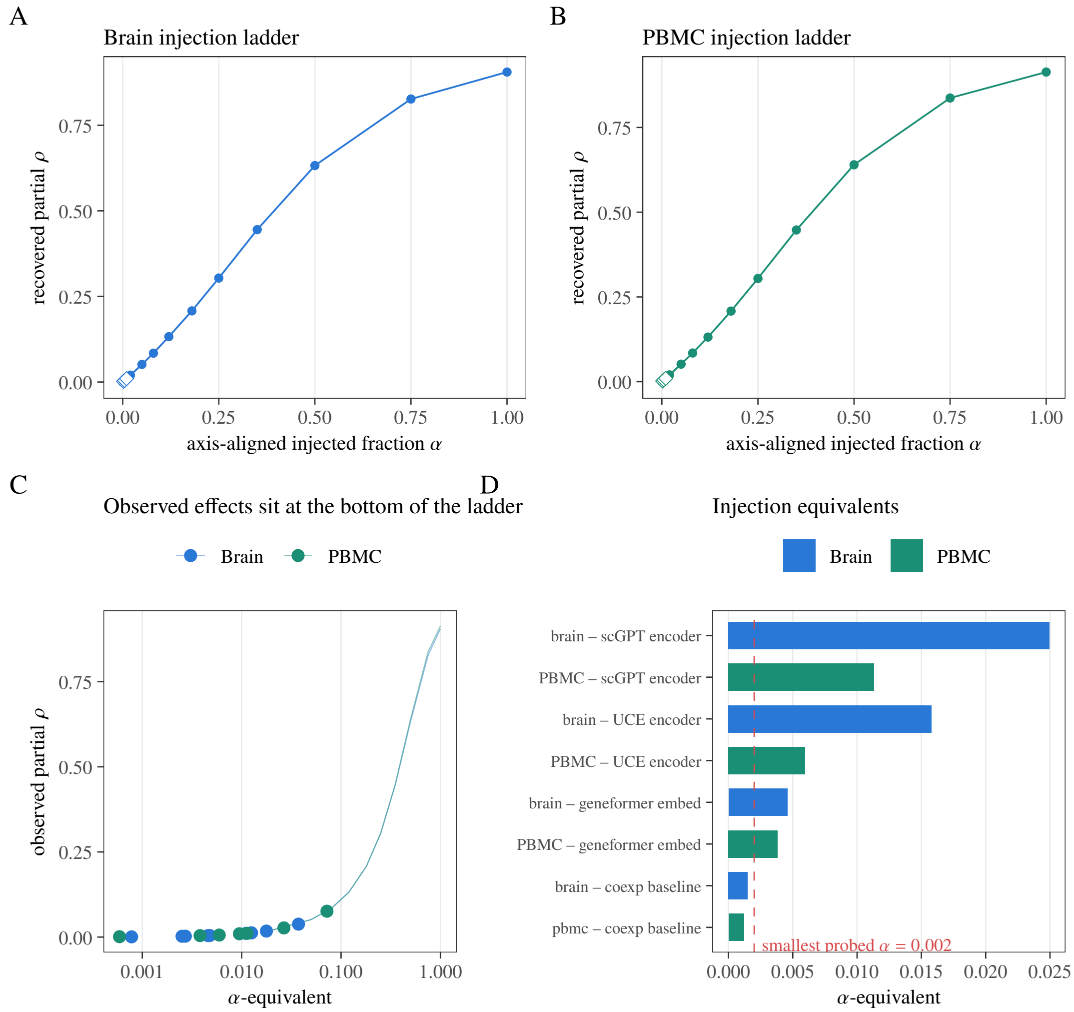

Figure 9. Axis-aligned pipeline-sensitivity diagnostic and observed-effect scale. (A–B) Brain and PBMC ladders; lines show the mean across 30 independent noise replicates, ribbons the replicate range, and diamonds the subdivided low-α points. (C) Observed effects located on the injection curves. (D) α-equivalents of the six supported positive rows relative to the smallest probed nonzero fraction; the negative brain attention row has no positive α-equivalent. This is not a confidence interval, power curve, MDE, or exclusion analysis.

models do not separate from the confound they are adjusted against. Second, the two supported readouts of the same model in the same tissue (Geneformer embedding $+0.0044$, Geneformer attention -0.0036, both brain) have opposite signs, which is not the signature of a model-level regulatory capability. Third, three of the six positives are negative before conditioning on the proxy's own degree covariates, so they appear only in the specification that conditions on functions of the outcome. A supervised TF-disjoint probe, answering the distinct question of supervised recoverability on paired PBMC data (the probe does not cover the brain rows, whose RNA and ATAC come from unpaired sources), finds mixed contrasts: UCE and the random-init floor are supported against co-expression ($q = 0.0325$ and $q = 0.0300$), while Geneformer embedding, Geneformer attention, and scGPT are not ($q = 0.5125$, $q = 0.65$, and $q = 0.255$). The random-init result cautions against reading the UCE contrast as model-specific regulatory recovery.

The most defensible reading is therefore not "four models encode weak regulation." It is that small, readout-specific, specification-dependent alignments exist on this fixed panel, that they are concentrated in embedding-geometry readouts rather than attention readouts, and that under this audit they cannot be distinguished from residual confound structure. A further observation points the same way: our own protocol pre-registers a readout sanity check in which a usable readout should correlate with co-expression (attention $\approx$ co-expression is well established). The supported positives come from readouts that fail

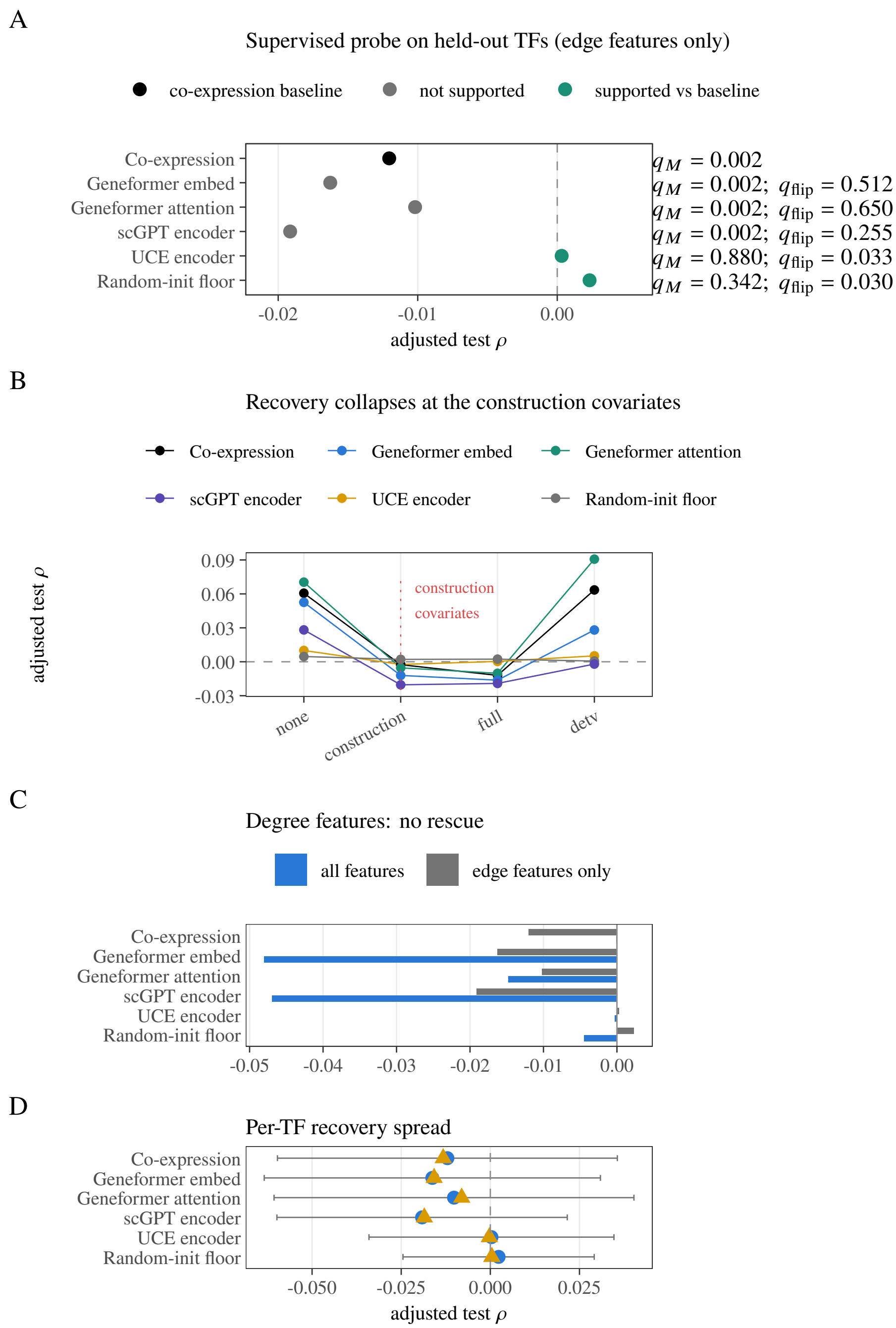

Figure 10. TF-disjoint supervised probe. (A) Confound-adjusted test correlation per family on 135 held-out TFs; labels give the gene-label permutation q_M and, for each FM family, paired sign-flip q_{flip} against co-expression. Green rows are supported against the baseline at $q_{\text{flip}} < 0.05$ (UCE 0.0325; random-init floor 0.0300). (B) Recovery across confound subsets. (C) Edge-only primary arm versus the all-feature sensitivity arm. (D) Across-TFmean, median, and one standard deviation of adjusted recovery.

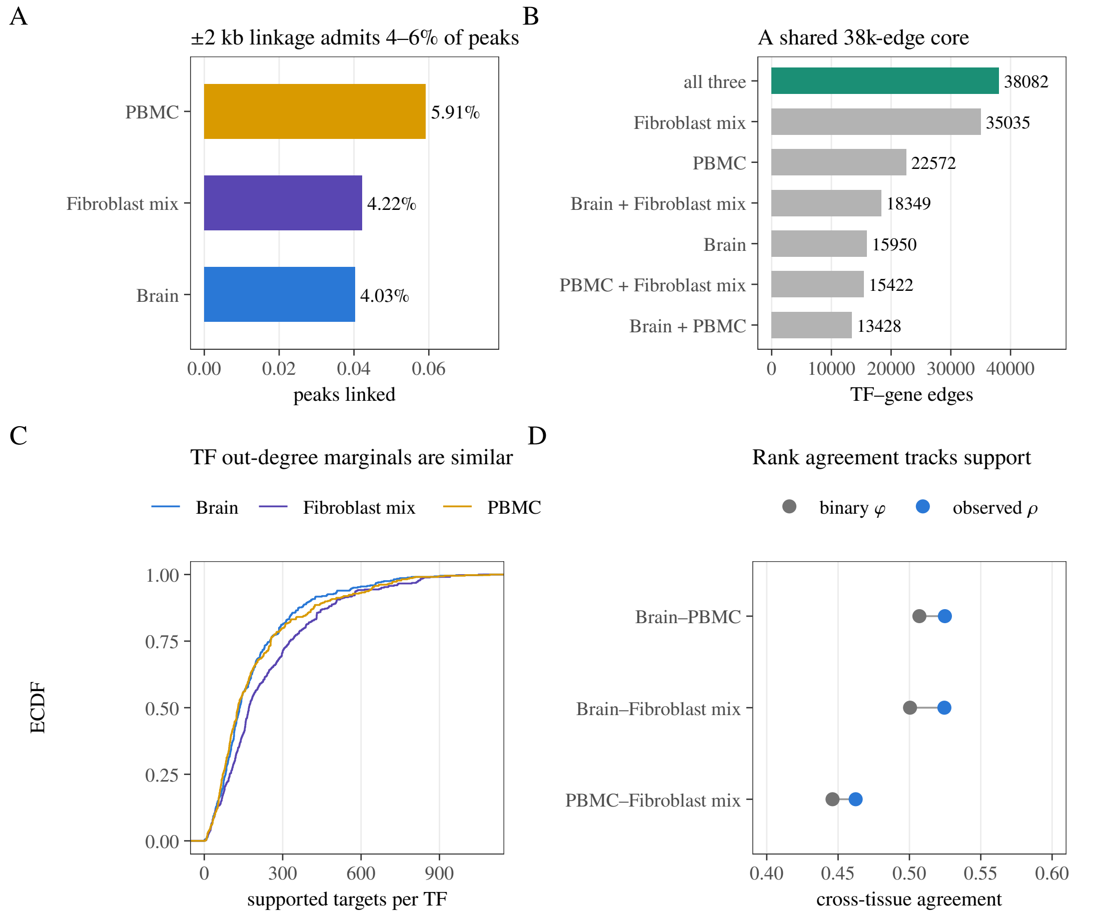

Figure 11. Third-tissue transfer of the proxy construct. (A) Fraction of each dataset's peaks admitted by the ±2 kb linkage window (relevant over total peaks). (B) Overlap of supported (nonzero) non-self TF–gene edges across the three consensus proxies; the green bar is the 38,082-edge core present in all three. (C) Empirical CDF of TF out-degree (supported targets per TF) per tissue. (D) Observed cross-tissue Spearman ρ versus binary-support φ for the three tissue pairs.

that check (FM–co-expression Spearman -0.18 to $+0.01$), while the readouts that pass it (scFoundation $+0.08$, PBMC Geneformer attention $+0.40$) are null. Embedding-cosine anisotropy interacting with degree conditioning is a more parsimonious account of that inversion than regulatory encoding. This is compatible with reports that scFM-derived gene relationships often resemble co-expression [13], and it does not license either a global regulatory-capability claim or a claim that FMs contain no regulatory information: the probe is linear and operates on pairwise graph summaries, and the proxy is itself a co-expression-family object.

Three statistical caveats apply throughout. First, one proxy randomization per replicate is shared across the rows of a family, so the Monte Carlo p-values within a family are positively correlated; Benjamini–Hochberg remains valid under positive regression dependence [5], but the family-level error rate is then a joint property of the shared randomization—rejections co-occur across rows that share replicates, so the guarantee is weaker than the independent-per-row reading, not stronger. Second, in the non-degree specification the row-shuffle null preserves degree while the statistic does not condition on it, so the two nulls ask different questions there: several non-degree rows are unusual only under the row shuffle (brain Geneformer attention at $q_D = 0.003$ and the random-init floor at $q_D = 0.045$), which we read as evidence that degree adjustment—not regulatory content—drives that null's rejections. Third, replication depth differs by model: scGPT is supported in both tissues but not by the probe; UCE is supported in both tissues but its probe contrast depends on the family boundary; Geneformer is supported only through one of two readouts and its other readout is opposite-signed. We therefore report readout-level, not model-level, support.

Limitations and scope. The reference is a regulatory-potential proxy built from motif and accessibility evidence, and its coverage is narrow by construction. The ± 2 kb promoter-plus-gene-body rule retains 6,609 of 111,743 accessible peaks in the PBMC data (5.91%); distal enhancer elements, which carry most cell-type-specific TF occupancy [25], are systematically excluded, and no co-accessibility or 3D-genome linker (for example Cicero [19] or activity-by-contact models [9]) is used. At the MOODS threshold used here ($p < 10^{-5}$, 500 bp accessible windows, both strands, 554 motifs), the expected random hit count is 5.5 per peak while the observed counts are 20.5 (brain), 23.2 (PBMC), and 21.7 (fibroblast mix), so roughly a quarter of motif hits are expected false positives [26]; no information-content threshold, conservation filter, or PWM-length correction [23] is applied. The panel is also composition-biased: 30 TFs in brain (31 in PBMC) share an identical binary target profile with at least one other TF (five FOX-family members are one such group), 128 of 446 TFs in brain (140 in PBMC) have a partner at binary cosine > 0.8, and in the reference lineage lists we checked, 7 of 21 neural TFs are present while all 37 housekeeping TFs are, because the panel cap ranks candidates by RNA detection rate. The proxy is near cell-type-invariant on this panel (pairwise $\rho \approx 0.99$), so per-cell-type rows score readouts against a reference with essentially no cell-type contrast. Brain analyses combine unpaired snATAC-seq with cross-study RNA, whereas the PBMC data are paired multiome, so the co-expression control is intrinsically stronger in PBMC than in brain. The full specification conditions on proxy-derived degree covariates; the non-degree co-primary specification avoids that hazard and supports no row under both nulls. The panel is fixed rather than sampled, so p_{MC} describes unusualness under the stated randomizations, not uncertainty over a TF or tissue population; no effect size carries an interval estimate, and with $N \approx 5.3 \times 10^5$ edges, effects of $|\rho| \approx 0.002$ are detectable, so statistical support alone cannot distinguish a small effect from no biologically meaningful effect. With $N = 999$ permutations, 0.001 is the resolution floor of every Monte Carlo p-value, not an exact value; rows with zero null exceedances are reported as $p < 0.001$. The supervised TF-disjoint probe is run on paired PBMC multiome data only; it does not directly test the eight brain rows, whose RNA and ATAC measurements come from unpaired sources. Per-cell-type and cross-tissue analyses are descriptive. The injection diagnostic is axis-aligned and cannot establish power against arbitrary biological alternatives. Fine-tuning, few-shot adaptation, decoder outputs, non-linear probes on raw per-gene embeddings, other gene panels, and direct perturbational validation are not tested. To make the path to broader validation explicit, we have inventoried and registered two additional single-cell RNA collections for follow-up replication: a development collection (27 datasets, $\approx 2.35 \times 10^6$ cells, mixed human and mouse) and a cancer collection (28 datasets, $\approx 1.20 \times 10^6$ cells, predominantly human), with per-file accession, cell count, and modality recorded in the project registry. These collections are RNA-only, so they cannot drive the accessibility/motif proxy used here; they are reserved for a planned co-expression-side replication that reuses the same fixed panel and confound specification once an orthogonal binding reference (for example ENCODE TF ChIP-seq) is in place.

5 CONCLUSION

scReg-Eval provides a reproducible fixed-panel protocol for separating co-expression and structural covariates from alignment with an accessibility/motif regulatory-potential proxy. Applied to thirteen pooled model/readout rows, it finds six small positive and one negative alignment supported by two randomization nulls, a co-expression baseline that reaches the same order of alignment, sign-inconsistent readouts within the same model, and positives that depend on conditioning on outcome-derived covariates. A supervised disjoint-TF probe gives mixed family-level contrasts, with the UCE and random-init floors distinguishing from co-expression but no model-specific regulatory recovery. The result motivates readout-specific, null-explicit, baseline-reported accounting rather than broad claims that scFMs either do or do not encode gene regulation.

DATA AND CODE AVAILABILITY

The fixed manifest, graph-construction and statistical code, deterministic seed metadata, complete model/readout coverage table, authoritative JSON outputs, and figure generator are provided in the project repository at `https://github.com/PeterPonyu/scfm-reg-audit` and archived at `https://doi.org/10.5281/zenodo.21724336`. Reproducibility is offered at three explicit layers: (i) artifact validation (the audit capsule's `validate_artifacts.py` verifies every published number and hash without downloads); (ii) statistical reruns of the fixed-panel audit against the released JSONs; and (iii) a full-pipeline recipe with dataset accessions, checkpoint SHA-256s, a locked environment, and ordered commands (`docs/FULL_RERUN.md` in the repository). Public inputs include GSE174367 brain data [17], the 10x Genomics 10k PBMC multiome [1], and the GSE206767 chemical-reprogramming ATAC pool. Large upstream datasets and model weights are not redistributed; the availability record documents their stable source identifiers and the exact manifests used. GSE206767 is a chemical-reprogramming experiment: it contains day-0 BJ fibroblasts and day-3 induced neuron-like cells, pooled here as one accessibility track because no per-barcode condition metadata is distributed with the matrix [10].

ETHICS STATEMENT

This study reuses public, de-identified datasets and performs no new human-subject recruitment or intervention.

COMPETING INTERESTS

The author declares no competing interests.

FUNDING AND AUTHOR CONTRIBUTIONS

This work received no dedicated or external funding.

CRediT author contributions: Zeyu Fu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing—original draft, Writing—review and editing, and Project administration.

DECLARATION OF GENERATIVE AI USE

A large language model (Anthropic Claude, accessed through an agentic coding assistant) was used to assist engineering and refactoring of the analysis code, hardening of the statistical pipeline, copy-editing of the manuscript, and generation of the figure-generation and packaging scripts, including an internal simulated-review process used for quality control. No AI system generated the scientific data, the randomization outputs, or any reported number; every artifact was recomputed by the deterministic pipelines described in the Methods. The author reviewed and edited all AI-assisted outputs and takes full responsibility for the accuracy and integrity of the manuscript and the accompanying code. No AI tool is listed as an author. A disclosure package with representative before/after code and the prompts used is provided with the submission.

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